



LIGHTWEIGHT MULTIMODAL FUSION FOR REAL-TIME CYBERSECURITY OF OPTICAL AND THERMAL IN-SITU MONITORING SIGNALS IN METAL ADDITIVE MANUFACTURING

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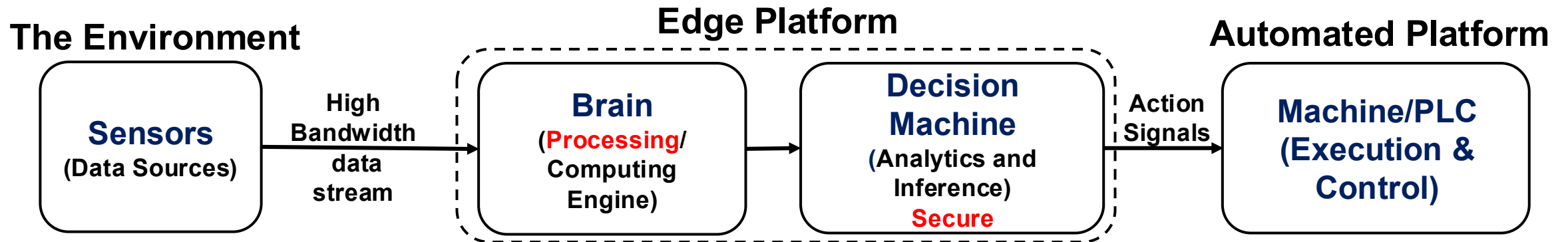
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Edge-Class Computing Platforms in Manufacturing Systems

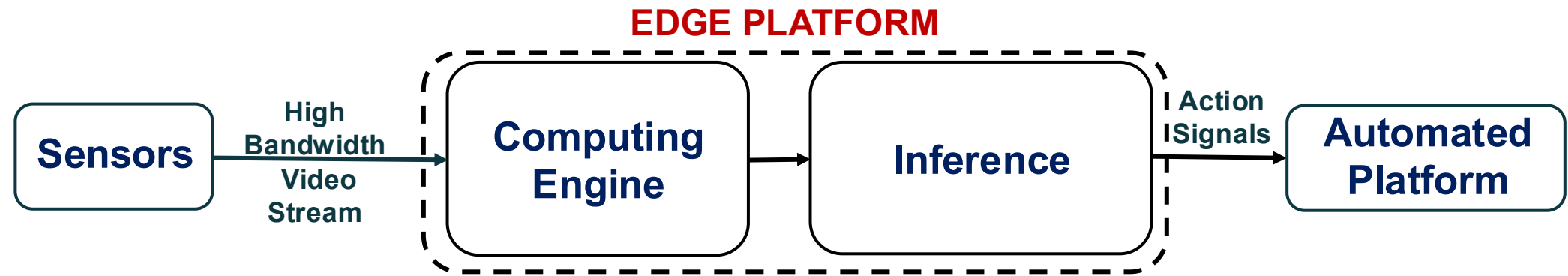
Edge Platforms: Acquire, Process and Secure High-Bandwidth Sensor Streams



Edge Platform

Edge platform consists of two main components:

Computing Engine and **Inference/Analysis Engine**

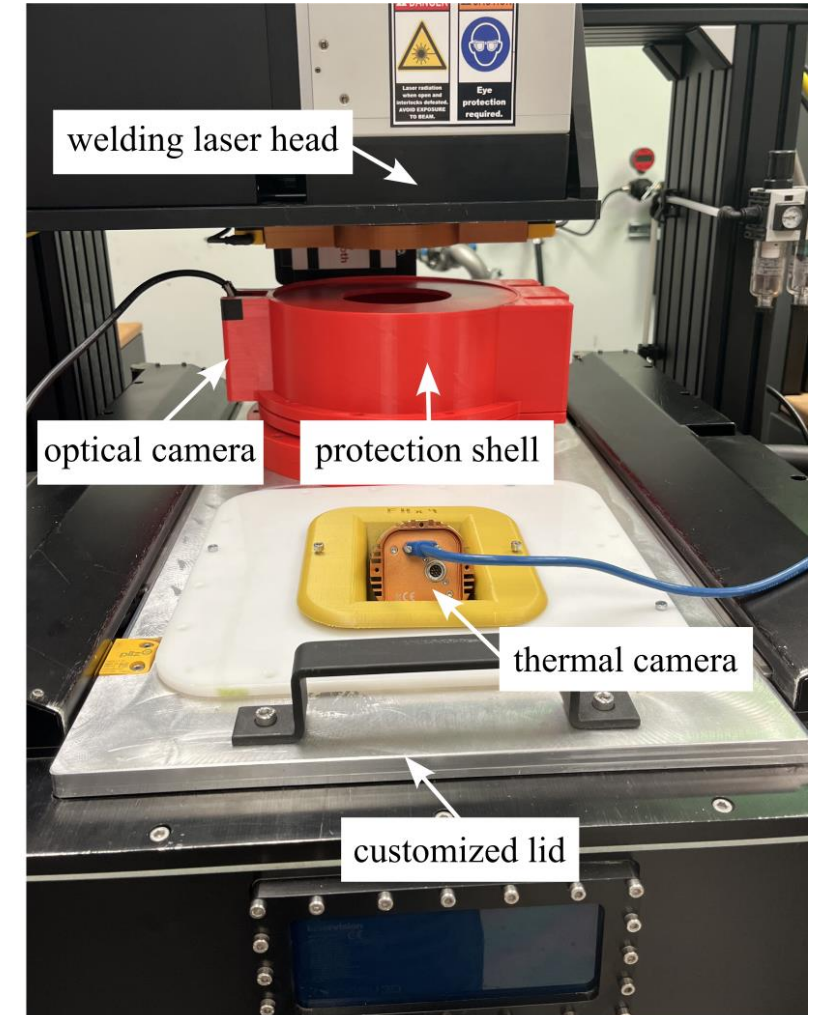


Edge Makes Decision Locally & Enable Real-Time Protection



EXAMPLE: BATTERY TAB LASER WELDING

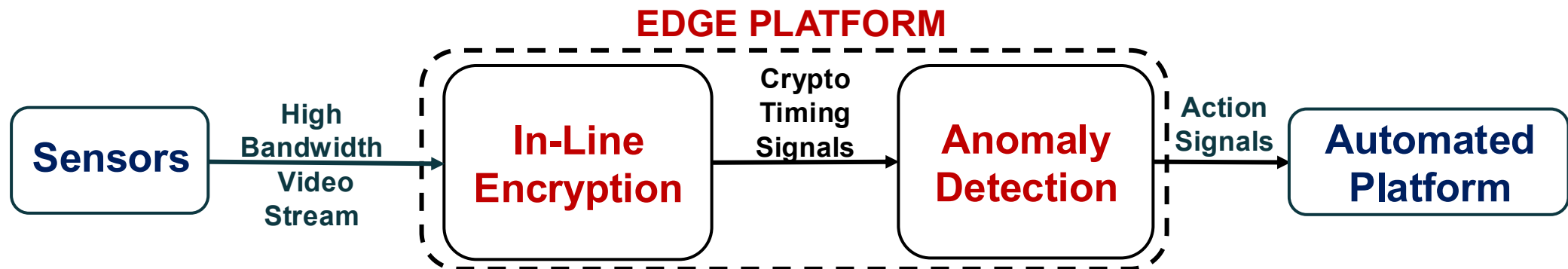
- **In-Situ Thermal Camera**
 - Monitors the weld region during laser welding
- **Process Monitoring**
 - Captures melt-pool behavior, heat distribution and cooling response
- **Quality Assessment**
 - Thermal features can support weld-quality classification



Experimental setup for in-situ optical and thermal monitoring of battery-tab laser welding (Hoang et al.)

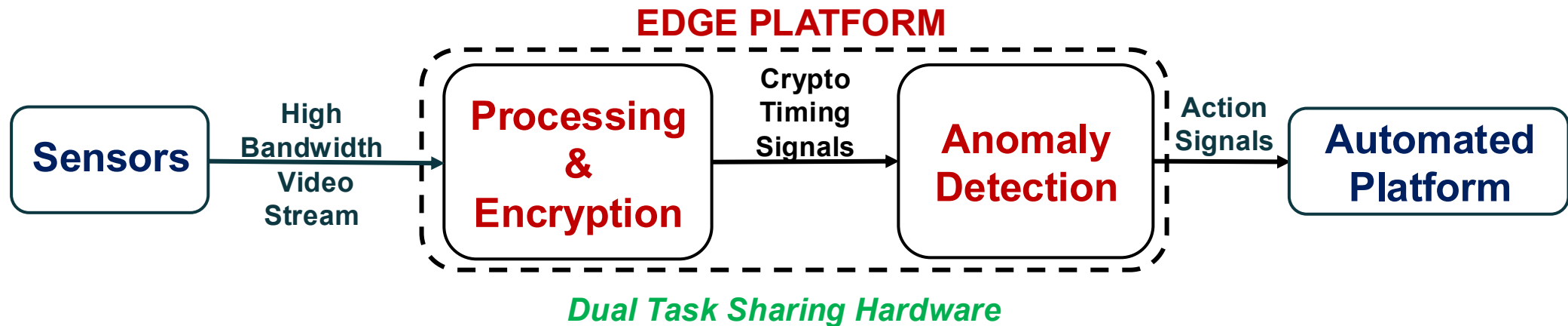
Edge Platform for Monitoring and Security: **Advantage**

Performing encryption and anomaly detection directly at the edge
reduces latency, preserves data privacy, and enables rapid mitigation of defects
before the next layer printed



Edge Platform for Monitoring and Security: **Disadvantage**

Embedding security into process monitoring create a key challenge:
Normal process variations and cyber manipulation can produce indistinguishable signatures in monitored signals.



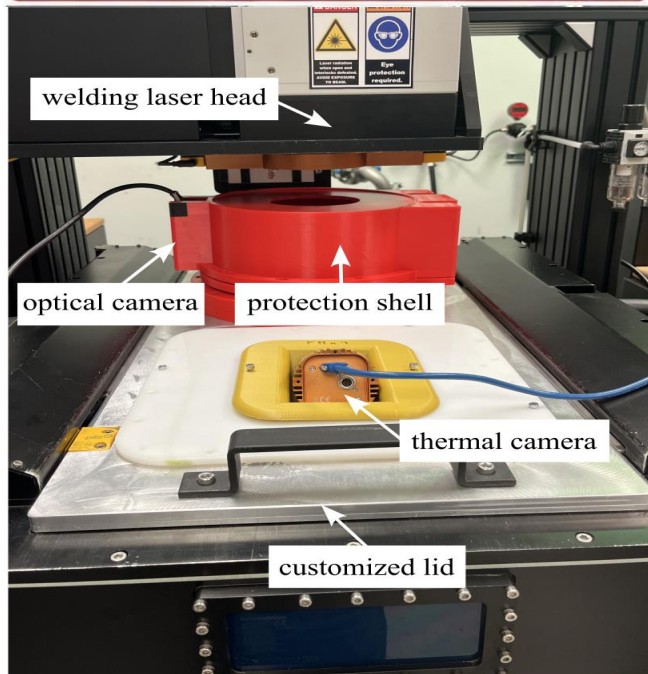
Shared Edge Hardware Creates a Noise-vs.-Attack Detection Challenge



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Attack/Malicious Manipulation Scenario

1. Normal Welding Process (Baseline Operation)



Possible Attacks

- Timing interference (execution, jitter)
- Fault injection (sensor/actuator)
- Power / voltage disturbance
- Electromagnetic disturbance
- Environmental manipulation

2. Attack-Induced Perturbations (Effects on System)

Timing Interference



Perturbs clock / execution timing

Fault Injection



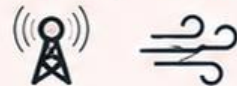
Induces transient faults or timing errors

Power / Voltage Disturbance



Causes power instability and timing variations

Electromagnetic Disturbance



Disrupts sensors and electronic signals

Environmental Manipulation



Alters temperature / EMI conditions

3. Changes in Monitored Signals (Attack Signatures)

Optical Signal

Intensity



Time

Abnormal intensity fluctuations

Thermal Signal

Temperature



Time

Unexpected thermal response

Timing (Execution) Signal

Execution Time



Time

Execution time variations / jitter

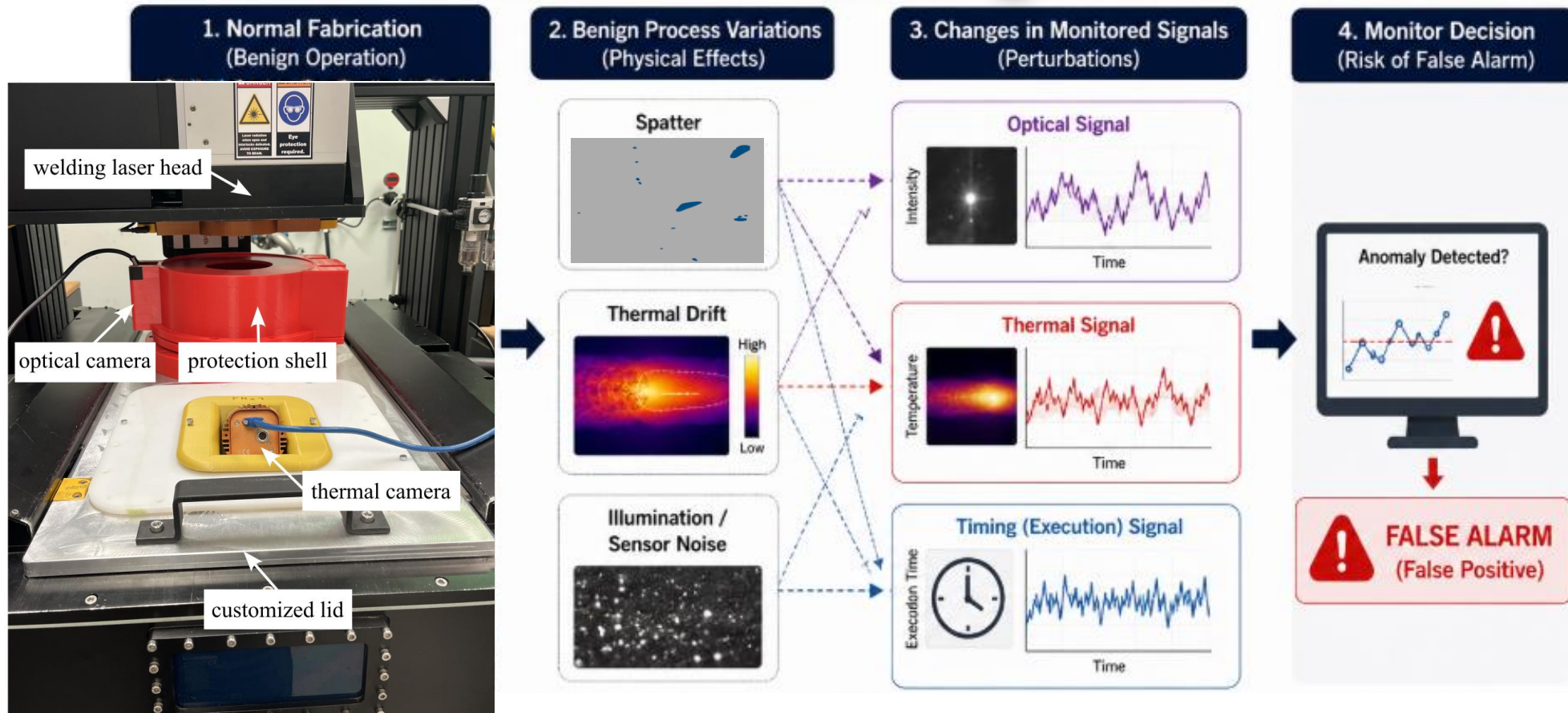
4. Monitor Design (Detection Challenge)

Anomaly Detected?



Attack detected as anomaly

Normal Process Variation (Noise) in Manufacturing



Normal process variations can perturb monitored signals and be misclassified as anomalies



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Prior Work

Prior work is effective but either resource-intensive or too dependent on timing, creating ambiguity between normal process variations and attacks.

Approach	Method	Strength / Purpose	Main Limitation
Traditional Methods (Protection)	Constant-time implementation	Prevents timing leakage	Performance & implementation overhead
	Shielding	Protects against physical/EM interference	Added cost & hardware complexity
	Hardware redundancy	Detects/tolerates injected faults	High area & power overhead
Timing-Only Monitoring	Runtime timing analysis	Lightweight & real-time	Cannot distinguish process variations from attacks

Research Gap: Lightweight monitoring is needed that can distinguish **benign process variations** from **malicious perturbations**.

What We Propose:

Multimodal Process-Aware Monitoring

Feature	Our Approach
Multimodal sensing	Timing + Optical + Thermal
Process awareness	Captures physical process context
Detection	Statistical + lightweight ML
Edge deployment	Low-latency, deterministic monitoring
Goal	Distinguish benign process variations from malicious attacks

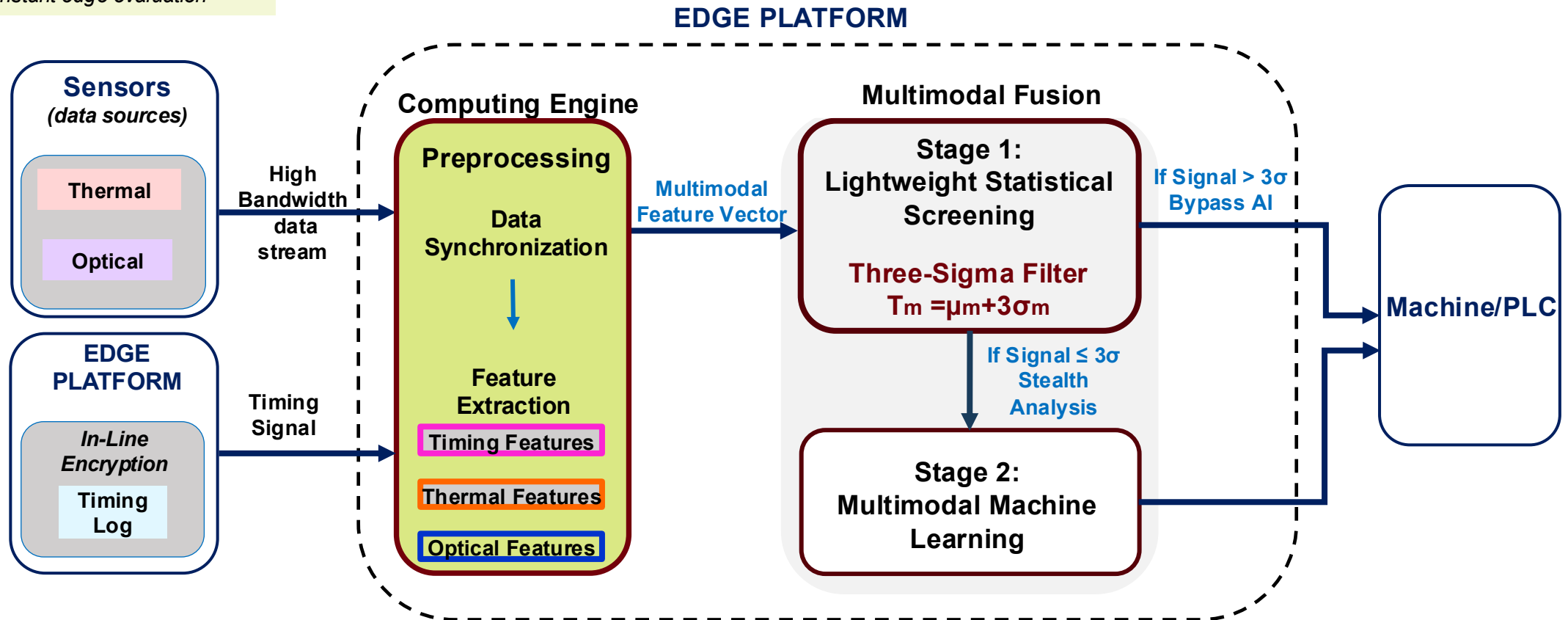
Move beyond timing-only monitoring by incorporating physical process context.



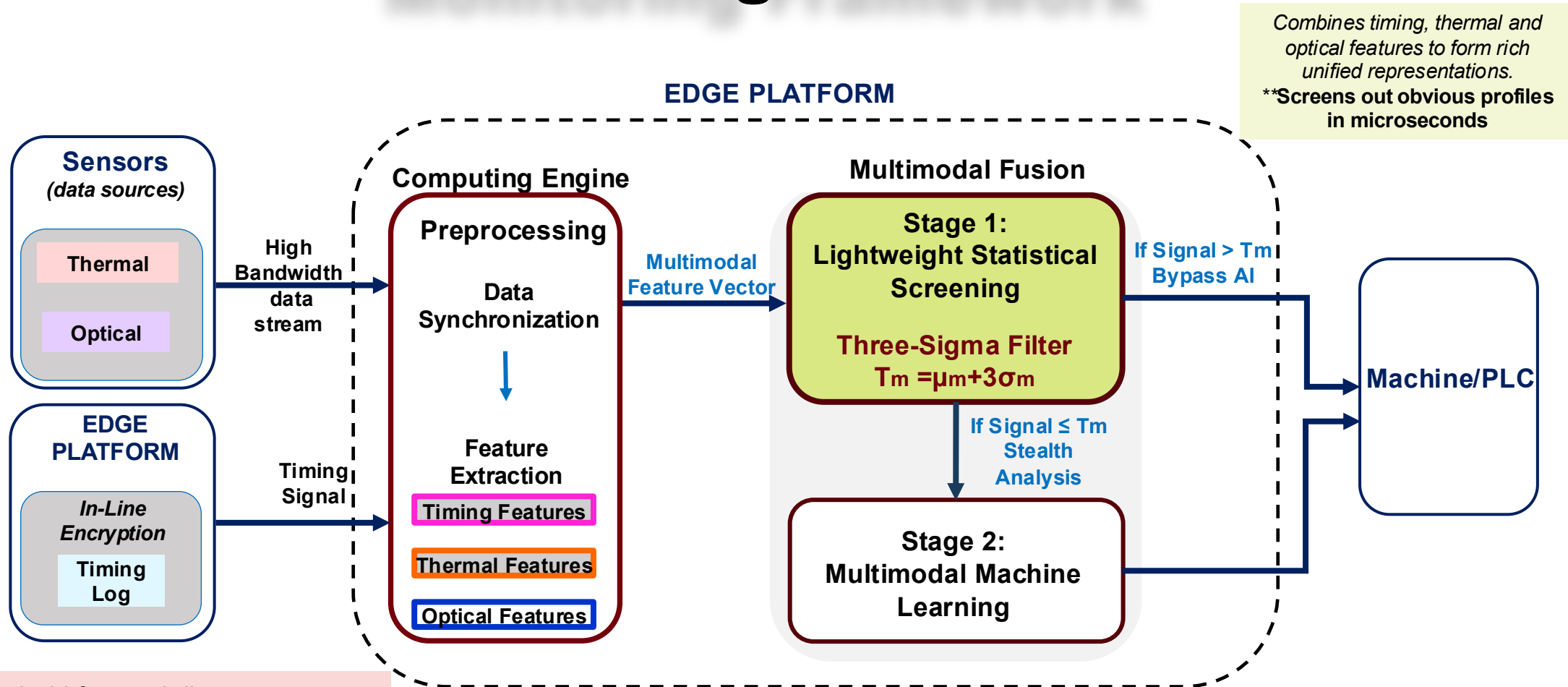
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Proposed Multimodal Process-Aware Monitoring Framework

Aligns inputs using shared microsecond timestamps, and compresses them into a lightweight three-element feature vector suitable for instant edge evaluation



Proposed Multimodal Process-Aware Monitoring Framework

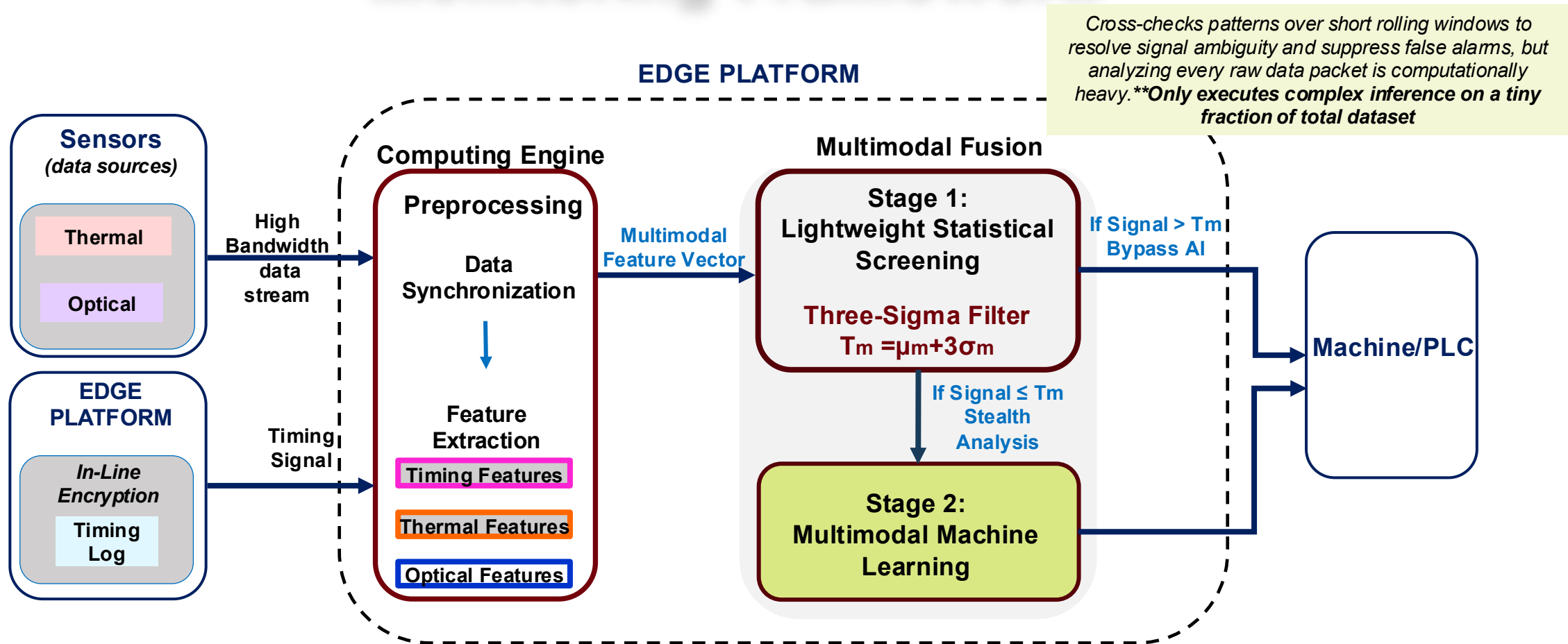


T_m – Threshold for modality m
 μ_m – Mean value for modality m
 σ_m – Standard deviation for modality m
 m – Modality index (time, therm, etc.)



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Proposed Multimodal Process-Aware Monitoring Framework



Evaluation

➤ Objective

- Compare timing-only and multimodal monitoring for distinguishing normal process variability from malicious anomalies.

➤ Dataset

- 12 manufacturing datasets containing synchronized timing, thermal, and optical measurements.
- Datasets span different materials, welding conditions, and data volumes.
- Data collected under controlled fabrication conditions.

➤ Data Modalities

- **Timing:** execution-time / timing traces from the embedded platform.
- **Thermal:** infrared measurements of the melt-pool / process region.
- **Optical:** process images capturing melt-pool and surface behavior.

➤ Dataset Categories

- Normal process operation
- Benign process variations
(e.g., *spatter, thermal fluctuations, illumination/sensor noise*)
- Malicious / attack-induced perturbations



Evaluation Dataset

12 datasets spanning different materials, weld conditions, and data volumes were used to evaluate accuracy, latency, and throughput.

Dataset Group	Examples	Size Range
Welding	Tab Welding, Laser Welds, Weld 01/02	0.54–100.59 MB
Steel	Steel No Glass 1/2, Thermal Test Weld Steel 1/2	3.48–115.47 MB
Aluminum	Thermal Test Weld Aluminum	125.17 MB
Battery	Battery Tab Weld	52.60 MB
Other Tests	Test Weld Recording, Filter Weld 2	0.54–96.62 MB

Experimental datasets provided by the ARTS Laboratory, University of South Carolina.

In-Situ Monitoring of Powder Bed Fusion Additive Manufacturing dataset repository

github.com/ARTS-Laboratory/In-situ-monitoring-of-powder-bed-fusion-additive-manufacturing

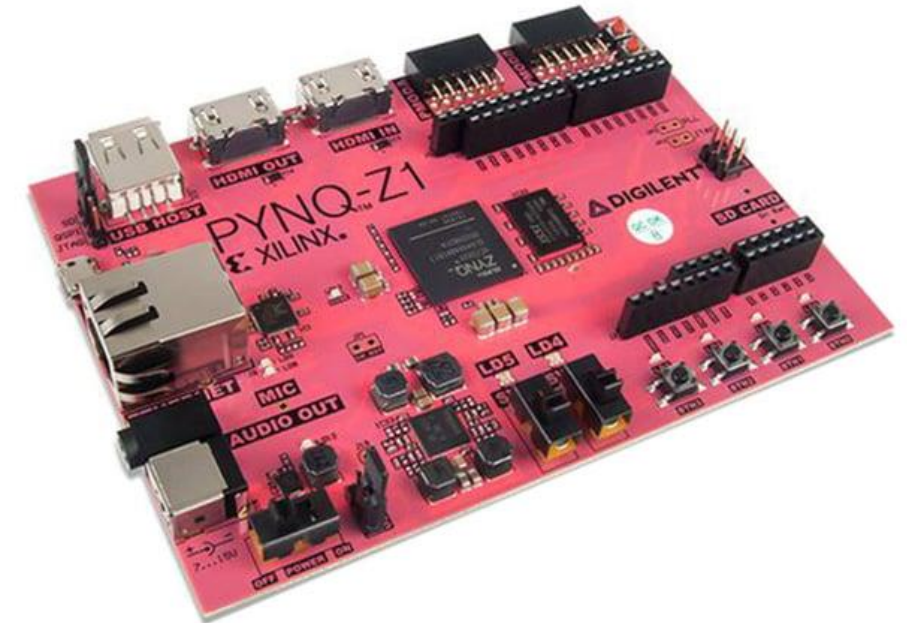


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Hardware Deployment Specification

Processor & Hardware Architecture

- Target Platform: PYNQ-Z2 edge development board.
- Evaluation CPU: Embedded ARM Cortex-A9 processor.
- Hardware Fabric: Programmable Logic (FPGA) fabric co-processor.
- Clock Frequency: Constrained at a stable 100 MHz.
- Total LUTs: 53,200
- Total FFs: 106,400
- Total BRAMs: 4.9 Mbits
- Total DSP Slices: 220 MAC



Evaluation on Traditional Embedded CPU

-36% Latency Reduction

Metric	Timing-Only	Full Hybrid (Optical + Thermal + Timing)
Accuracy	~99.9%	~99.9%
Inference Latency	~1.42 ms	~0.91 ms
Throughput	Dataset-dependent	Comparable / slightly higher
Multimodal Context	No	Yes
Process-Aware Detection	Limited	Improved

Timing-only: CPU must run heavy ML inference on **100% of packets** to check for leaks.
Full Hybrid: Stage 1 gatekeeper: Uses fast, constant-time statistics across all channels.
Short-circuit evaluation: Massive anomalies bypass Stage 2 (ML) instantly, saving CPU cycles.

Confusion Matrix Comparison

The hybrid fusion model eliminates false negatives across all evaluated datasets, improving anomaly sensitivity, but introduces a higher false-positive rate in some cases.

Metric	Timing-Only	Full Hybrid (O + T + Time)	Observation
False Negatives (FN)	43	0	No missed anomalies
False Positives (FP)	23	58	More conservative detection
True Negatives (TN)	High	Higher	Strong normal-case recognition
True Positives (TP)	High	High	Maintains strong attack detection



Localized Thermal Drift Anomalies

94% of Excess FPs Confined to Only 2 Datasets

Dataset	Timing-Only FP	Hybrid FP
Thermal Test Weld Steel 1	2	23
Thermal Test Weld Steel 2	1	15

- Thermal welding signals naturally vary because of temperature changes and cooling cycles.
- Stage 1 quickly detects deviations across data channels.
- In volatile thermal-steel data, normal shifts may be flagged as anomalies, causing false positives before ML analysis



Hardware Deployment Results

Metric	Result	Significance
FPGA LUT Utilization	155 LUTs (0.29%)	Very compact hardware footprint
Flip-Flop Utilization	129 FFs (0.12%)	Minimal logic overhead
FPGA Clock	100 MHz	Deterministic operation
FPGA Latency	10 ns/block	1 block/cycle
HW Throughput	1600 MB/s	Theoretical high-throughput processing
Measured ARM Inference	897–933 μ s/block	Software/I/O overhead
Memory Usage	9.60–9.79 MB	Low and stable
Power	0.141 W total	Low power for continuous edge operation
Dynamic Power	0.036 W	Low processing overhead

The system performs real-time edge inference in under **1 ms per block** while using less than **10 MB of memory**. FPGA processing reaches **1600 MB/s**, far exceeding ARM throughput.

Edge Runtime Performance

Metric	Result
Inference Latency	897–933 μ s/block
Total Processing Time per Dataset	24.98–25.79 s
ARM Throughput	0.04–9.27 MB/s
FPGA Throughput	1600 MB/s
Memory Usage	9.60–9.79 MB

- The system performs real-time edge inference in under **1 ms per block** while using less than **10 MB of memory**.
- FPGA processing reaches **1600 MB/s**, far exceeding ARM throughput.



Conclusion

❖ Multimodal monitoring improves process awareness

Combined **timing, optical, and thermal signals** provide richer process context than timing-only monitoring.

❖ Lightweight fusion enables real-time edge monitoring

The hybrid approach maintained **~99.9% accuracy** while reducing inference latency from **1.42 ms to 0.91 ms (~36%)**.

❖ Physical context improves anomaly sensitivity

The hybrid detector reduced **false negatives from 43 to 0** but increased false positives in some thermally variable datasets.

Cybersecurity monitoring in manufacturing should consider both cyber signals and physical process behavior.



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Future Work

- ❖ **Reduce false positives** under sustained thermal/process variation.
- ❖ **Expand multimodal datasets** and process conditions.
- ❖ **Further optimize edge implementation** for practical real-time deployment



Thank You!

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